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Transition control of a tail-sitter unmanned aerial vehicle with L1 neural network adaptive control

Jingyang ZHONG^{*}, Chen WANG, Hang ZHANG

School of Construction Machinery, Chang'an University, Xi'an 710061, China

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Transition strategy

Abstract The main task of this work is to design a control system for a small tail-sitter Unmanned Aerial Vehicle (UAV) during the transition process. Although reasonable control performance can be obtained through a well-tuned single PID or cascade PID control architecture under nominal conditions, large or fast time-varying disturbances and a wide range of changes in the equilibrium point bring nonlinear characteristics to the transition control during the transition process, which leads to control precision degradation. Meanwhile, the PID controller's tuning method relies on engineering experiences to a certain extent and the controller parameters need to be retuned under different working conditions, which limits the rapid deployment and preliminary validation. Based on the above issues, a novel control architecture of L1 neural network adaptive control associated with PID control is proposed to improve the compensation ability during the transition process and guarantee the security transition. The L1 neural network adaptive control is revised to solve the multi-input and multi-output problem of the tail-sitter UAV system in this study. Finally, the transition characteristics of the time setting difference between the desired transition speed and the desired transition pitch angle are analyzed.

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1. Introduction

In recent years, with the development of sensors and unmanned technology, small Unmanned Aerial Vehicles (UAVs) have constantly attracted researchers' attention world-

wide^{1,2}. Regardless of civil fields, including cargo transportation, pipeline patrol and forest rescue, or military fields, including mission reconnaissance and target attack, small UAVs all play important roles in helping people improve work efficiency and mitigate risks^{3–5}.

Small tail-sitter UAVs can both hover and rapidly fly forward, leading to rapid deployment ability under different working conditions, which makes them one of the most promising types of unmanned aerial vehicles⁶. In recent years, researchers and institutions from various countries have always maintained a high degree of enthusiasm for tail-sitter UAV related research. Olsson et al.⁷ focused on aerodynamic coefficient identification over the entire flight envelope of a

^{*} Corresponding author.

E-mail address: zhongjy@chd.edu.cn (J. ZHONG).

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WingtraOne tail-sitter UAV, and the results showed that the model was well suited for predicting the forces and moments. Raj et al.⁸ studied a real-time algorithm to iteratively learn a forward transition maneuver via repeated flight trials for a biplane-quadrotor tail-sitter UAV. Li et al.⁹ focused on transition process optimization for a quad-rotor tail-sitter UAV, and both simulation and experimental results showed that the optimized transition trajectory could ensure a short transition time and small height increment. Swarnkar et al.¹⁰ developed a control strategy for a novel biplane-quadrotor tail-sitter UAV and the numerical simulations verified the good controller performance. Manzoor et al.¹¹ used offset-free model predictive control to a ducted fan tail-sitter UAV and simulations were computed to investigate the control performance. Furthermore, a ducted fan tail-sitter UAV called V-bat has been studied by researchers from America and Korea since 2011. Related papers, patents and flight experiments have always been tracked in recent years^{12–14}. All flight experiments are reported to be conducted in naval ships by U.S. military up to September 2021, which fully illustrates the much attention given to the tail-sitter UAV field. Furthermore, researchers from different countries and organizations, such as Gill and D'Andrea¹⁵, Smeur et al.¹⁶ and Ritz and D'Andrea¹⁷ all spread out the study in this field.

The most commonly used control architecture for tail-sitter UAVs is the PID controller. Although reasonable control effects can be obtained under general working conditions, the linear foundation of the PID controller may lead to control precision degradation when the tail-sitter UAV encounters large or fast time-varying disturbances¹⁸. A wide range of changes in the equilibrium point during the transition process also brings nonlinear characteristics that further increase the control difficulty which is disadvantageous for transition safety. Furthermore, to obtain a satisfactory control effect, the controller parameters need to be retuned according to different working conditions. The parameter tuning method also relies on engineering experiences to a certain extent¹⁹ and a matter of trial and error has to be conducted to obtain ideal parameters.

The L1 adaptive control architecture^{20,21}, proposed by Chengyu Cao and Naira Hovakimyan, can effectively compensate for the system's uncertainties and improve control precision. L1 adaptive control grows out of the Model Reference Adaptive Control (MRAC), and the low-pass filter that L1 adaptive control brings into the system input makes the system performance not approach an ideal system, but a reference system²². This low-pass filter makes the stability and performance analysis different between L1 adaptive control and model reference adaptive control. L1 adaptive control only compensates for uncertainties within the control bandwidth, which could avoid the robustness decrease in fast adaptation. However, the standard MRAC does not possess this characteristic and the robustness degrades with fast adaptation.

From the research results in recent years, L1 adaptive control has been investigated in aeronautical fields. Lu and Liu²³ took the UAV carrier landing control problem as the research object, and nonlinear dynamic inversion and L1 adaptive control methods were used to deal with the disturbance during landing. Jafarnejadsani et al.²⁴ focused their research on the optimal low-pass filter design of L1 adaptive control and verified the design results in the trajectory tracking for

a small quadrotor UAV. Ackerman et al.²⁵ presented the evaluation results of an L1 stability augmentation system implemented on a variable-stability Learjet aircraft for the first time and the results illustrate that the L1 stability augmentation system could significantly restore the dynamic characteristics and provide consistent and safe handling qualities in nonlinear working conditions. Banerjee et al.²⁶ studied the lateral/directional maneuver of a hypersonic glider along a descent trajectory. A pole placement controller augmented with an L1 adaptive controller is designed and verified through simulations on attitude and altitude control subjects. The architectures of L1 adaptive control theory have guaranteed performance and guaranteed robustness in the presence of fast adaptation, without introducing or enforcing persistence of excitation, without any gain scheduling in the controller parameters and without resorting to high-gain feedback. The speed of adaptation in L1 adaptive architectures is limited only by the available hardware, while robustness is resolved via conventional methods from classical and robust control and this is the advantage compared to the general adaptive or robust control in the real flight.

Although L1 adaptive control has been studied in aeronautical fields, there still exist problems worth further exploration. First, few researchers consider using an L1 adaptive controller to augment the PID controller for tail-sitter transition flight control. The control system that we design here can make use of L1 adaptive control's fast compensation ability to cover the PID control shortage and accelerate the deployment of the PID controller under practical circumstances. Second, Radial Basis Function Networks (RBFNs) have the characteristics of simple construction, strong nonlinear approximation ability, and good extensibility and classification ability, which have received attention from researchers in different fields. However, most researchers consider uncertainties to be linear functions of system states, and the combination between L1 adaptive control and RBFN is rarely studied. Although Wang et al.²⁷ considered the combination of L1 adaptive control and RBFN to solve the autonomous aerial refueling problem, the control architecture that Wang et al.²⁷ mentioned could only deal with the Single-Input Multi-Output (SIMO) problem. Therefore, Wang et al.²⁷ divided the original problem into multiple SIMO problems for consideration. Finally, most researchers study the tail-sitter UAV transition process based on the optimal idea, and the altitude restriction is always added to the objective function. Few researchers have investigated the effects of the relationship between the time of the desired pitch angle and the time of the desired flight speed on the transition characteristics under a linear transition strategy.

Based on the above analysis, this study focuses on tail-sitter UAVs' attitude control during the transition process. The main contribution of this work is given as follows. (A) A complete dynamic model of a dual-rotor tail-sitter UAV is built and detailed features are considered. (B) A control architecture of L1 neural network adaptive control associated with PID control is proposed to address fast time-varying disturbances, a wide range of equilibrium point changes and uncertainties during the transition process. (C) L1 neural network adaptive control is improved to address Multi-Input Multi-Output (MIMO) problems, which has not been done before and the corresponding theories are revised. (D) The influences of the time setting between the desired pitch angle and desired flight

speed on the transition characteristics are quantitatively analyzed, and initial conclusions are given.

2. System description and dynamics

To ensure the rationality and reliability of tail-sitter UAV transition control analysis, the system description and modeling process are given first.

2.1. System description

The tail-sitter UAV in this study is shown in Fig. 1, and the dual-rotor flying-wing configuration is adopted. In Fig. 1, T is the thrust generated by two motors and propellers, ΔT_1 and ΔT_2 are the thrust variations generated by two motors for rolling control.

The coordination system's definition is also illustrated in Fig. 1. During the transition process, the thrusts generated by two motors and propellers are used to accelerate the UAV to reach the transition finish speed as quickly as possible. Furthermore, the differential thrusts generated by two propellers are also used to provide the rolling control moment. The symmetry and asymmetry deflection of the control surfaces are used for pitching and yawing control, and the slipstream generated by propellers has a critical impact on the efficiency of the control surfaces.

2.2. System conceptual and aerodynamic parameters

The tail-sitter UAV conceptual parameters in this study are shown in Table 1.

For the real UAV in this study, the maximal control surface deflection is approximately $\pm 25^\circ$. To test the control performance under tougher simulation conditions, $\pm 20^\circ$ is set here.

The transition process mainly occurs in the X_b - Z_b coordinate plane, and lateral/directional motion is small. Lateral/directional aerodynamics show a weak correlation with the angle of attack. Therefore, longitudinal aerodynamic characteristics are mainly considered and the data are provided by Sun et al.²⁸ as shown in Fig. 2²⁸.

The remaining aerodynamic parameters were obtained by vortex lattice method from our previous studies and are shown in Table 2.

2.3. System kinematics and dynamics

The tail-sitter UAV is regarded as a rigid body, and then the system kinematics and dynamics can be defined as

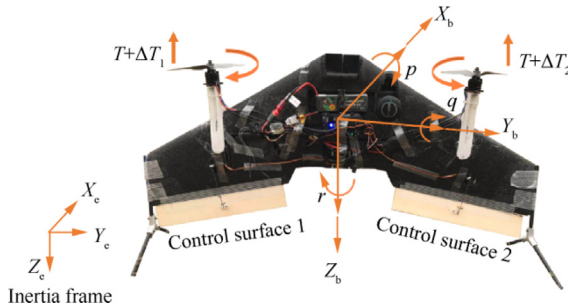


Fig. 1 Prototype of tail-sitter UAV.

Table 1 Tail-sitter UAV's conceptual parameters.

Symbol of parameter	Description	Value
m (kg)	Mass	0.81
c (m)	Reference chord	0.253
b (m)	Reference span	0.8774
δ_c ($^\circ$)	Maximal acceptable control surface deflection	± 20
J (kg·m ²)	Inertia matrix	$[0.025, 0, 0; 0, 0.007, 0; 0, 0, 0.022]$
C_{cs} (m)	Control surface chord	0.05
C_{ss} (m)	Control surface span	0.3
T (N)	Maximal acceptable motor thrust	6.25

$$\dot{\Omega} = R\omega \quad (1)$$

$$R = \begin{bmatrix} 1 & \sin \phi \tan \theta & \cos \phi \tan \theta \\ 0 & \cos \phi & -\sin \phi \\ 0 & \sin \phi \sec \theta & \cos \phi \sec \theta \end{bmatrix} \quad (2)$$

$$\dot{\omega} = -B(\omega \times J\omega) + BM \quad (3)$$

where $\Omega = [\varphi, \theta, \psi]$ is the Euler angle; $\omega = [p, q, r]$ is the angular rate; $B = J^{-1}$ is the inversion of the moment of inertia; M is the moment acting on the tail-sitter UAV and has five parts.

$$M = M_{\text{tor}} + M_{\text{aero}} + M_{\text{en}} + M_c + M_{\text{unk}} \quad (4)$$

where M_{tor} is the propeller torsion moment; M_{aero} is the aerodynamic moment; M_{en} is the environmental disturbance moment; M_c is control moment generated by actuators; M_{unk} is unmodeled moment.

2.3.1. Motor characteristics

The torsion moment and thrust generated by the propeller can be modeled by

$$M_{\text{tor}} = k_M \Omega_i^2 \quad (5)$$

$$T = k_T \Omega_i^2 \quad (6)$$

where Ω_i is the motor speed; k_M and k_T are the propeller torsion coefficient and force coefficient respectively. k_M and k_T are obtained through experiments and the values of them are 6.618×10^{-8} and 1.35×10^{-9} . The motor models are mainly used for control allocation. Although a nonlinear relationship between the square of motor speed and thrust or torque would better fit the experiment data, this would add complexity when calculating motor speed from thrust and torque. The coefficient of determination R^2 for linear fit between the square of motor speed and thrust and torque in Tables 3 and 4 are 0.9997 and 0.9967, respectively, which means that the linear fit is pretty good. The experimental data, in which the thrusts and torques are calculated according to the sensor voltages, are listed in Table 3 and Table 4.

The tail-sitter UAV's propellers are mounted on the leading edge of the wing, the aerodynamic characteristics were

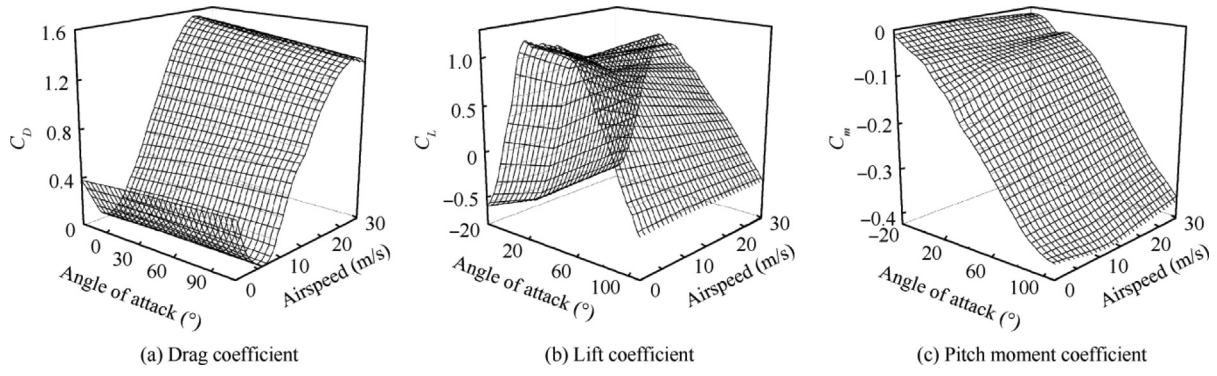


Fig. 2 Static longitudinal aerodynamic coefficients of tail-sitter UAV.²⁸

Table 2 Tail-sitter UAV's aerodynamic coefficients.

Symbol of parameter	Description	Value
C_{mq}	Pitch damping coefficient	-1.01
C_{lp}	Roll damping coefficient	-0.016
C_{nr}	Yaw damping coefficient	-0.327
$C_{l\beta}$	Lateral stability derivative	-0.0001
C_{l0}	Roll moment coefficient at zero angle of attack	-0.00005
C_{n0}	Yaw moment coefficient at zero angle of attack	-0.00003
C_{lr}	Roll cross derivative	0.026
$C_{n\beta}$	Directional stability derivative	-0.00003
C_{np}	Yaw cross derivative	0.024
$C_{m\delta e}$	Pitch control derivative	-0.2857
$C_{n\delta e}$	Yaw control derivative	0.1562

Table 4 Torque experiment data.

Motor speed (r/min)	PWM signal (μ s)	Torque (N·m)
3380	1252	0.0146
4540	1304	0.0226
6040	1390	0.0426
7130	1469	0.0626
7780	1519	0.0767
8570	1591	0.0927
9900	1720	0.1307

$$V_{\text{avg}}(x) = V_{\text{avg0}} \left[1 + \frac{x/R_p}{\sqrt{1 + (x/R_p)^2}} \right] \quad (7)$$

$$R_s(x) = R_p \sqrt{\frac{V_{\text{avg0}}}{V_{\text{avg}}(x)}} \quad (8)$$

where R_p is the propeller-disc radius; V_{avg0} is the induced velocity near the propeller disc ($x = 0$); $V_{\text{avg}}(x)$ is the slipstream speed behind the propeller disc at a distance of x ; $R_s(x)$ is the slipstream radius behind the propeller disc at a distance of x . V_{avg0} can be obtained as

$$T = 2\rho A(w + U_{\text{gust}} + V_{\text{avg0}})V_{\text{avg0}} \quad (9)$$

where T is the propeller thrust; ρ is the air density; A is the propeller-disc area; w and U_{gust} are the flight speed and environment disturbance speed in the x -axis of the body coordinate, respectively.

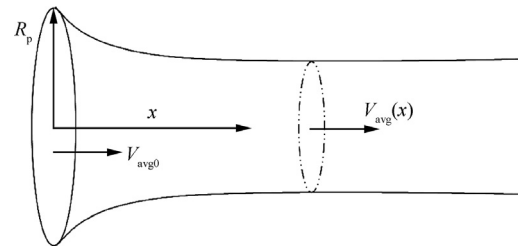


Fig. 3 Propeller slipstream (momentum theory).

influenced by the slipstream propeller to a large extent and they are modeled as follows.

2.3.2. Propeller slipstream

The slipstream speed and slipstream area can be obtained through momentum theory.²⁹

The dynamic pressure and areas in different areas, which are within and outside the propeller slipstream, can be calculated based on the above equations (see Fig. 3).

2.3.3. Dynamic pressure and angle of attacks in different areas

The propeller slipstream influences the dynamic pressure by changing the speed acting on the wing, which leads to different aerodynamic characteristics. In addition, speed changes also lead to variation in the angle of attack, which impacts aerodynamic coefficients. Therefore, the angles of attack in different areas are illustrated in Fig. 4.

The angles of attack within and outside the propeller slipstream can be obtained as

$$\alpha_{in} = \arctan\left(\frac{u}{w + V_{slip}}\right) \quad (10)$$

$$\alpha_{out} = \arctan\left(\frac{u}{w}\right) \quad (11)$$

where u and w are flight speeds in body coordinates; α_{out} is the angle of attack outside the propeller slipstream; α_{in} is angle of attack within propeller slipstream; V_{slip} is slipstream speed generated from propeller.

2.3.4. Tail-sitter UAV dynamic characteristics

Considering that the wing and control surfaces are divided into different areas within and outside the propeller slipstream as shown in Fig. 5, it is necessary to grasp these different characteristics. That means the aerodynamic force and moment should be considered separately in different areas. The detailed modeling process is given below.

$$\mathbf{f} = \mathbf{f}_{in} + \mathbf{f}_{out} \quad (12)$$

$$\mathbf{f}_* = \mathbf{R}_{e(*)}^b \mathbf{F}_* \quad (13)$$

$$\mathbf{F}_* = [D_* \quad Y_* \quad L_*] \quad (14)$$

$$\mathbf{f} = [f_x \quad f_y \quad f_z] \quad (15)$$

$$\mathbf{f}_{in} = [f_{xin} \quad f_{yin} \quad f_{zin}] \quad (16)$$

$$\mathbf{f}_{out} = [f_{xout} \quad f_{yout} \quad f_{zout}] \quad (17)$$

$$\mathbf{R}_{e(*)}^b = \begin{bmatrix} \cos(\alpha_*) \cos(\beta) & -\cos(\alpha_*) \sin(\beta) & -\sin(\alpha_*) \\ \sin(\beta) & \cos(\beta) & 0 \\ \sin(\alpha_*) \cos(\beta) & -\sin(\alpha_*) \sin(\beta) & \cos(\alpha_*) \end{bmatrix} \quad (18)$$

$$L_* = 0.5 C_L(\alpha_*) \rho V_*^2 S_* \quad (19)$$

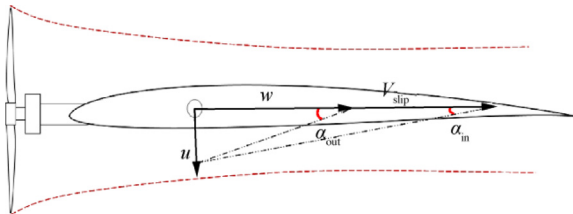


Fig. 4 Angle of attack in different areas.

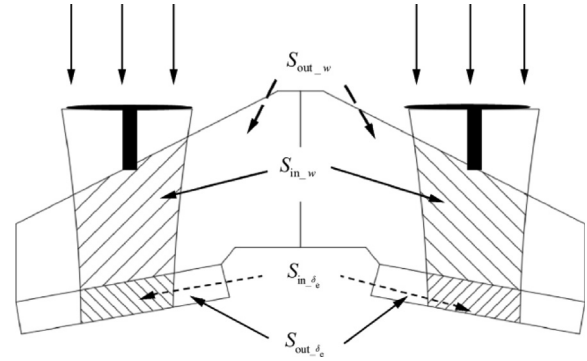


Fig. 5 Influence of propeller slipstream.

$$Y_* = 0.5 C_Y(\beta) \rho V_*^2 S_* \quad (20)$$

$$D_* = 0.5 C_D(\alpha_*) \rho V_*^2 S_* \quad (21)$$

$$S_* = S_{*\delta_e} + S_{*w} \quad (22)$$

where $\mathbf{f}$ is the total force acting on UAV, f_x , f_y and f_z are the force acting on the UAV's X_b axis, Y_b axis and Z_b axis, respectively, f_{xin} , f_{yin} , f_{zin} , f_{xout} , f_{yout} and f_{zout} are drag, side force and lift within and outside propeller slipstream in body coordinate, respectively, α_* represents the angle of attack, when “*” is taken for “in”, it means that the state to be described is within the propeller slipstream, and when “*” is taken for “out”, it means that the state to be described is outside the propeller slipstream, $\mathbf{R}_{e(*)}^b$ is the coordinate transformation matrix from body coordinates to wind coordinates, D_* , Y_* and L_* are drag, side force and lift in wind coordinate, respectively, $S_{*\delta_e}$ and S_{*w} are the dynamic areas of the control surfaces and wing where dynamic means the area changes with the propeller thrust, V_* is the sum of airspeed and slipstream speed near the quarter chord of the wing, ρ is air density.

Aerodynamic moments are shown as Eqs. (23)–(33).

$$\mathbf{M}_{aero} = \mathbf{M}_{aero_in} + \mathbf{M}_{aero_out} \quad (23)$$

$$\mathbf{M}_{aero} = [M_{aero(l)} \quad M_{aero(m)} \quad M_{aero(n)}] \quad (24)$$

$$\mathbf{M}_{aero*} = [M_{aero(l)*} \quad M_{aero(m)*} \quad M_{aero(n)*}] \quad (25)$$

$$M_{aero(m)*} = 0.5 \rho V_*^2 S_* c C_{m*}(\alpha_*) \quad (26)$$

$$M_{aero(l)*} = 0.5 \rho V_*^2 S_* b C_l(\beta) \quad (27)$$

$$M_{aero(n)*} = 0.5 \rho V_*^2 S_* b C_n(\beta) \quad (28)$$

$$M_{\delta_r} = M_{\delta_r_in} + M_{\delta_r_out} \quad (29)$$

$$M_{\delta_e} = M_{\delta_e_in} + M_{\delta_e_out} \quad (30)$$

$$M_{\delta_a} = (\Delta T_1 - \Delta T_2) l \quad (31)$$

$$M_{\delta_r*} = 0.5 \rho V_{\delta_e*}^2 S_* b C_{m\delta_a} \delta_a \quad (32)$$

$$M_{\delta_e*} = 0.5 \rho V_{\delta_e*}^2 S_* c C_{m\delta_e} \delta_e \quad (33)$$

where $\mathbf{M}_{aero}$ is the total moment acting on UAV, $M_{aero(l)}$, $M_{aero(m)}$, $M_{aero(n)}$ are the moment acting on the UAV's X_b axis,

Y_b axis and Z_b axis, respectively, M_{δ_a} is the roll moment generated by motor and propeller, M_{δ_r} and M_{δ_e} are the yaw moment and pitch moment generated by control surface, respectively, c and b are reference chord and span, respectively, l is the distance between central axis of motor and Z_b axis of the body, $V_{\delta_{e-*}}$ is the sum of airspeed and slipstream speed near the quarter chord of the control surface.

2.3.5. Environment disturbance

Environmental disturbances are mainly caused by wind. According to the MIL-F-885C standard, wind can be divided into three different forms: gust, turbulence and wind shear³⁰. The gust is illustrated here. Furthermore, turbulence and wind shear could also be modeled according to the MIL-F-885C standard which is not shown here.

$$V_{\text{wind}} = \begin{cases} 0 & x < 0 \\ \frac{V_m}{2} \left[1 - \cos\left(\frac{\pi x}{d_m}\right) \right] & 0 \leq x \leq d_m \\ V_m & x > d_m \end{cases} \quad (34)$$

where V_m is gust amplitude; d_m is gust length; V_{wind} is wind in body coordinates.

The model established above is mainly used for flight simulation and controller verification, and the following controller design also refers to the established model.

3. Control system design

The cascade PID controller is designed first for the tail-sitter UAV transition process. However, the nature of linear control for PID control would lead to performance degradation under large and fast time-varying nonlinear disturbances. The L1 neural network adaptive control is then designed for compensation to improve the tail-sitter UAV's transition safety.

3.1. Cascade PID control

The cascade PID control consists of an angle loop and an angular rate loop. Eq. (1) constitutes the kinematical equation of the angle loop, which involves the coordinate transformation. Eq. (3) represents the angular rate dynamics. From the different forms of moment acting on the UAV in Eq. (4), the disturbances and uncertainties are all reflected in the angular rate loop. Since Eq. (1) is essentially a coordinate transformation equation, there are no uncertainties and unmodeled factors. Therefore, the first loop of cascade PID does not need to use the PID structure but the P control alone. Eq. (3) contains various sources of the moment, including aerodynamic moment, wind field disturbing moment, unmodeled error, control moment of actuators, and other factors. Integration is

needed to compensate for these uncertainties and disturbances, so the final cascade PID uses the form P + PID to offer basic transition performance. The controller's architecture and implementation details are given as Fig. 6.

$$\Omega_e = (\Omega_d - \Omega) \quad (35)$$

$$\omega_d = P_{\Omega} \Omega_e \quad (36)$$

$$\omega_e = \omega_d - \omega \quad (37)$$

$$u = P_{\omega} \omega_e + I_{\omega} \int \omega_e dt + D_{\omega} \frac{a}{s+a} \dot{\omega}_e \quad (38)$$

where Ω_d are desired Euler angles that UAV should follow; Ω_e are errors between desired and actual Euler angles; ω_d are desired angular rates; ω_e are errors between desired and actual angular rates; $a/(s+a)$ is the low-pass filter and a is the bandwidth of the low-pass filter.

Then, L1 neural network adaptive control is designed for better compensation for disturbances and uncertainties.

3.2. L1 neural network adaptive control

3.2.1. Basic control architecture

In this section, a nonlinear controller is designed to compensate for the environmental disturbances and unmodeled uncertainties mentioned above. From the previous analysis, it is known that disturbances and uncertainties are mainly reflected in Eq. (3) and Eq. (4), therefore, the following dynamics are used for controller design:

$$\dot{\omega} = -B(\omega \times J\omega) + B(M_{\text{tor}} + M_{\text{aero}} + M_{\text{dis}} + M_c + M_{\text{unk}}) \quad (39)$$

where M_c is the control moment that the UAV's actuators should generate, and the desired values of the control moment are computed from the control algorithm.

Eq. (39) is then reformed as

$$\dot{\omega} = Bu_{L1} + B(M_{\text{tor}} + M_{\text{aero}} + M_{\text{dis}} + M_{\text{unk}} - \omega \times J\omega) \quad (40)$$

A new variable $f(\omega, t)$ is defined to represent the items within brackets on the right-hand side of Eq. (40) and M_c is replaced with u_{L1} , which leads to the following equation:

$$\dot{\omega} = B(u_{L1} - f(\omega, t)) \quad (41)$$

where $B \in \mathbb{R}^{3 \times 3}$ is the inversion of the triaxial moment of inertia, $\omega \in \mathbb{R}^3$ is the UAV angular rate, $u_{L1} \in \mathbb{R}^3$ is the control moment generated by the L1 neural network adaptive controller, $f(\omega, t) : \mathbb{R}^3 \times \mathbb{R} \rightarrow \mathbb{R}^3$ is the total disturbances and uncertainties.

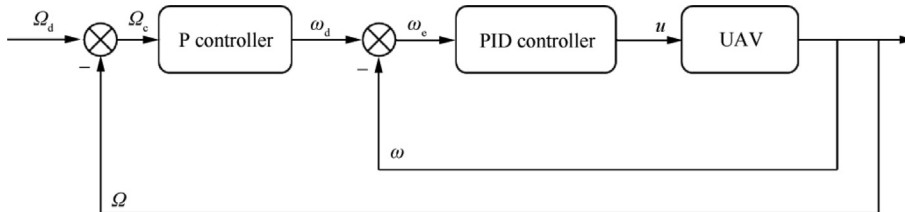


Fig. 6 Cascade PID controller architecture.

The reason why we use $\mathbf{f}(\boldsymbol{\omega}, t)$ to represent the items within brackets on the right-hand side of Eq. (40) is presented as follows: The model errors between simulation and practical UAV systems are inevitable. The design ideas mentioned above can reduce the dependence of the control algorithm on the model accuracy. The effect of the design idea can be verified through simulation in section 4 because the controller design process only uses part of the model information while the simulation environment is a complete nonlinear model. If the controller could not cope with unknown uncertainties and disturbances well, the attitude tracking performance would be poor.

By defining a desired angular rate dynamic that the L1 neural network adaptive controller should follow, Eq. (41) can be rewritten as

$$\dot{\boldsymbol{\omega}} = \mathbf{A}_m \boldsymbol{\omega} + \mathbf{B}(\mathbf{u}_{L1} - \mathbf{f}(\boldsymbol{\omega}, t)) \quad (42)$$

where $\mathbf{A}_m$ is a Hurwitz matrix, $\mathbf{f}(\boldsymbol{\omega}, t)$ is an unknown mapping that satisfies the Lipschitz continuity condition.

$$\|\mathbf{f}(t, 0)\|_{\infty} \leq \bar{B} \quad (43)$$

$$\|\mathbf{f}(t, \boldsymbol{\omega}_1) - \mathbf{f}(t, \boldsymbol{\omega}_2)\|_{L_{\infty}} \leq L \|\boldsymbol{\omega}_1 - \boldsymbol{\omega}_2\|_{L_{\infty}} \quad (44)$$

The control objective is to design an L1 neural network adaptive controller so that UAV angular rate $\boldsymbol{\omega}$ can follow a bounded continuous desired angular rate $\boldsymbol{\omega}_d$, as shown in Eq. (36).

The system disturbances and uncertainties $\mathbf{f}(\boldsymbol{\omega}, t)$ represent the three-axis disturbances and uncertainties acting on the tail-sitter UAV. The following RBF networks are designed to estimate these disturbances in different directions.

$$\mathbf{f}(t, \boldsymbol{\omega}) = \mathbf{W}(t)\mathbf{H}(\boldsymbol{\omega}(t)) + \boldsymbol{\varepsilon} \quad (45)$$

where $\boldsymbol{\omega}$ is the network input, $\mathbf{H}(\boldsymbol{\omega}(t)) = [h_1(\boldsymbol{\omega}(t)), h_2(\boldsymbol{\omega}(t)), \dots, h_j(\boldsymbol{\omega}(t))](j = 1, 2, \dots, n)$, $\boldsymbol{\varepsilon} = [\varepsilon_1, \varepsilon_2, \varepsilon_3]$ is the network approximation error and $\|\boldsymbol{\varepsilon}\|_{\infty} \leq \varepsilon^*$. $\mathbf{W}(t) = [W_{11}(t), W_{12}(t), \dots, W_{1k}(t); W_{21}(t), W_{22}(t), \dots, W_{2k}(t); W_{31}(t), W_{32}(t), \dots, W_{3k}(t)]$ ($k = 1, 2, \dots, n$) are the ideal weights.

$$h_j = \exp \left[\frac{-(\boldsymbol{\omega} - \mathbf{z}_j)^T (\boldsymbol{\omega} - \mathbf{z}_j)}{\delta_j^2} \right] \quad (46)$$

where $\mathbf{z}_j$ is the center vector of the Gaussian basis function, δ_j is the width of the Gaussian basis function. The state predictor can be defined, according to Eq. (42), as

$$\dot{\hat{\boldsymbol{\omega}}}(t) = \mathbf{A}_m \hat{\boldsymbol{\omega}}(t) + \mathbf{B} \left(\mathbf{u}_{L1}(t) - \hat{\mathbf{W}}(t)\mathbf{H}(\boldsymbol{\omega}(t)) \right) \quad (47)$$

The estimation of $\mathbf{W}(t)$ is defined as $\hat{\mathbf{W}}(t)$ which is computed as

$$\dot{\hat{\mathbf{W}}}(t) = \Gamma \text{Proj} \left(\mathbf{P}\mathbf{B}\tilde{\boldsymbol{\omega}}(t)\mathbf{H}^T(\boldsymbol{\omega}(t)), \hat{\mathbf{W}}(t) \right) \quad (48)$$

where $\tilde{\boldsymbol{\omega}}(t) = \hat{\boldsymbol{\omega}}(t) - \boldsymbol{\omega}(t)$ are angular rate errors. $\hat{\mathbf{W}}(0)$ are the initial values of neural network weight, $\Gamma = \Gamma_c \mathbf{I}_{n \times n} \in \mathbf{R}^{n \times n}$ is adaptative gain, $\text{Proj}(\cdot, \cdot)$ represents projection operator. $\mathbf{P} = \mathbf{P}^T > 0$ is the solution of the Lyapunov equation $\mathbf{A}_m^T \mathbf{P} + \mathbf{P}\mathbf{A}_m = -\mathbf{Q}$ and $\mathbf{Q}$ is a strictly positive definite matrix. In simulation, the term $\mathbf{P}\mathbf{B}\tilde{\boldsymbol{\omega}}(t)\mathbf{H}^T(\boldsymbol{\omega}(t))$ and $\hat{\mathbf{W}}(t)$ are reshaped from matrix to vectors first for projection operator calculation, and then reshaped to a $3 \times n$ matrix for subsequent process.

By defining $\hat{\boldsymbol{\eta}}(t) = \hat{\mathbf{W}}(t)\mathbf{H}(\boldsymbol{\omega}(t))$, the control law can be defined as

$$\mathbf{u}_{L1}(s) = \mathbf{C}(s)\hat{\boldsymbol{\eta}}(s) + \mathbf{K}_g \boldsymbol{\omega}_d(s) \quad (49)$$

where $\mathbf{u}_{L1}(s)$, $\hat{\boldsymbol{\eta}}(s)$ and $\boldsymbol{\omega}_d(s)$ are the Laplace transformations of $\mathbf{u}_{L1}(t)$, $\hat{\boldsymbol{\eta}}(t)$ and $\boldsymbol{\omega}_d(t)$, respectively, $\mathbf{C}(s)$ is the strictly positive definite low-pass filter with a low-frequency gain of 1, $\mathbf{K}_g = -1/(\mathbf{A}_m^{-1}\mathbf{B}) = \text{diag}(k_{g1}, k_{g2}, k_{g3})$ allows the system to track the constant signal without steady-state error. The complete control architecture is shown in Fig. 7.

3.2.2. Control stability and performance analysis

In this part, the control stability and performance are analyzed. Lemma 1 is given first below.

Lemma 1. ³¹ For a stable proper MIMO system $\bar{\mathbf{H}}(s)$ with input $\mathbf{r}(t) \in \mathbf{R}^m$ and output $\mathbf{x}(t) \in \mathbf{R}^n$, we have $\|\mathbf{x}_t\|_{L_{\infty}} \leq \|\bar{\mathbf{H}}(s)\|_{L1} \|\mathbf{r}_t\|_{L_{\infty}}, \forall t \geq 0$.

To guarantee that the designed L1 adaptive control system is stable, the following L1 condition should be satisfied:

$$\|\mathbf{G}(s)\|_{L1} L < 1 \quad (50)$$

$$\mathbf{G}(s) = \bar{\mathbf{H}}(s)(\mathbf{C}(s) - \mathbf{I}) \quad (51)$$

$$\bar{\mathbf{H}}(s) = (s\mathbf{I} - \mathbf{A}_m)^{-1}\mathbf{B} \quad (52)$$

The following variables are given first to be convenient for the subsequent formula derivations.

$$\gamma_r = \frac{(\|\bar{\mathbf{H}}(s)\|_{L1} \|\mathbf{K}_g \boldsymbol{\omega}_d\|_{L_{\infty}} + \|\mathbf{G}(s)\|_{L1} (\bar{B} + \varepsilon^*) + \|\bar{\mathbf{H}}(s)\|_{L1} \varepsilon^*)}{1 - \|\mathbf{G}(s)\|_{L1} L} \quad (53)$$

$$\gamma_0 = \sqrt{\frac{\lambda_{\max}(\mathbf{P}) \left(\frac{2\varepsilon^* \|\mathbf{P}\mathbf{B}\|_{\infty}}{\lambda_{\min}(\mathbf{Q})} \right)^2 + \frac{W_{\max}}{\lambda_{\min}(\mathbf{P})\Gamma_c}}{\lambda_{\min}(\mathbf{P})\Gamma_c}} \quad (54)$$

$$\gamma_1 = \frac{5\varepsilon^* \|\mathbf{G}(s)\|_{L1} + (1 + \|\mathbf{C}(s) - \mathbf{I}\|_{L1})\gamma_0 + \|\bar{\mathbf{H}}(s)\|_{L1} \varepsilon^*}{1 - \|\mathbf{G}(s)\|_{L1} L} \quad (55)$$

$$\gamma_2 = \|\mathbf{C}(s)/(\mathbf{c}_o^T \bar{\mathbf{H}}(s))\mathbf{c}_o^T\|_{L1} \gamma_0 + \|\mathbf{C}(s)\|_{L1} L \gamma_1 + 3\|\mathbf{C}(s)\|_{L1} \varepsilon^* \quad (56)$$

Compared to the MRAC algorithm, the L1 adaptive controller introduces a low-pass filter to the system input. This low-pass filter makes the stability analysis of the L1 adaptive controller not the same as that of the MARC algorithm. The following reference system should be given first.

Reference system

The reference system defines the best performance that the real system could approach or a nonadaptive version of the adaptive control system in Eq. (42) and Eq. (49). The following control law is used in the reference system:

$$\mathbf{u}_{L1\text{-ref}}(s) = \mathbf{C}(s)\boldsymbol{\eta}(s) + \mathbf{K}_g \boldsymbol{\omega}_d(s) \quad (57)$$

where $\boldsymbol{\eta}(s)$ represents the ideal $\mathbf{W}(s)\mathbf{H}(\boldsymbol{\omega}_{\text{ref}}(s))$. By using the control law in Eq. (57), the closed-loop reference system can be obtained as

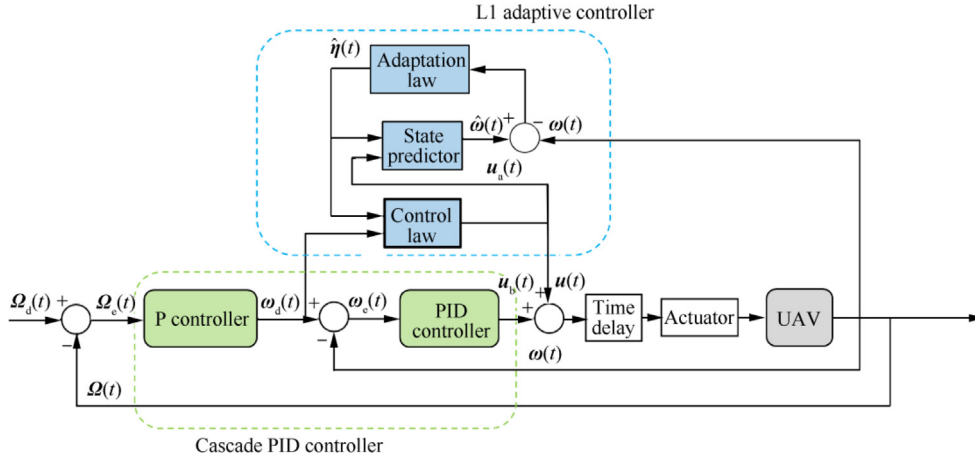


Fig. 7 Complete control architecture.

$$\begin{aligned} \omega_{\text{ref}}(s) = & \bar{H}(s)K_g\omega_d(s) + \bar{H}(s)(C(s) - I)W(s)H(\omega_{\text{ref}}(s)) \\ & - \bar{H}(s)\varepsilon(s) \end{aligned} \quad (58)$$

where $\omega_d(s)$, $W(s)H(\omega(s))$ and $\varepsilon(s)$ are the Laplace transformations of $\omega_d(t)$, $W(t)H(\omega(t))$ and $\varepsilon(t)$, respectively.

Lemma 2 states that the closed-loop system is stable and the state $\omega_{\text{ref}}(s)$ is bounded.

Lemma 2. The control signal given by Eq. (57), subject to the condition Eq. (50), ensures that the state of the closed-loop system Eq. (58) remains within the following bound:

$$\|\omega_{\text{ref}}\|_{L_\infty} \leq \gamma_r \quad (59)$$

Proof. Assume that the bound (59) does not hold. Then, since $\|\omega_{\text{ref}}(0)\|_\infty \leq \gamma_r$ and $\omega_{\text{ref}}(t)$ is continuous, there exists t' making $\|\omega_{\text{ref}}(t')\|_\infty > \gamma_r$ and $\|\omega_{\text{ref}}(\tau)\|_\infty \leq \gamma_r + \gamma_1$ for any $\tau \in [0, t']$, which leads to the following inequalities:

$$\gamma_r < \|\omega_{\text{ref},t'}\|_{L_\infty} \leq \gamma_r + \gamma_1 \quad (60)$$

Lemma 1 is applied to the closed-loop system in Eq. (58) which gives

$$\begin{aligned} \|\omega_{\text{ref},t'}\|_{L_\infty} & \leq \|G(s)\|_{L_1} \|\eta_{t'}\|_{L_\infty} \\ & + \|\bar{H}(s)\|_{L_1} \|K_g\omega_{d,t'}\|_{L_\infty} + \|\bar{H}(s)\|_{L_1} \varepsilon^* \end{aligned} \quad (61)$$

From Eq. (45), we know that $W^T h(\omega_{\text{ref}}(s)) = f(\omega_{\text{ref}}(s)) - \varepsilon(\omega_{\text{ref}}(s))$, and then from Eq. (43) and Eq. (44), we have

$$\|\eta_{t'}\|_{L_\infty} \leq \|f(t', 0)\|_\infty + L \|\omega_{\text{ref}}(t')\|_{L_\infty} + \varepsilon^* \quad (62)$$

Eq. (61) can then be rewritten as

$$\begin{aligned} \|\omega_{\text{ref},t'}\|_{L_\infty} & \leq \|G(s)\|_{L_1} \left(\|f(t', 0)\|_\infty + L \|\omega_{\text{ref},t'}\|_{L_\infty} + \varepsilon^* \right) \\ & + \|\bar{H}(s)\|_{L_1} \|K_g\omega_{d,t'}\|_{L_\infty} + \|\bar{H}(s)\|_{L_1} \varepsilon^* \end{aligned} \quad (63)$$

Using the condition (54), one gets

$$\|\omega_{\text{ref},t'}\|_{L_\infty} \leq \frac{(\|\bar{H}(s)\|_{L_1} \|K_g\omega_{d,t'}\|_{L_\infty} + \|G(s)\|_{L_1} (\bar{B} + \varepsilon^*) + \|\bar{H}(s)\|_{L_1} \varepsilon^*)}{1 - \|G(s)\|_{L_1} L} \quad (64)$$

Since $\|\omega_{d,t'}\| \leq \|\omega_d\|_{L_\infty}$, it is straightforward to see that $\|\omega_{\text{ref},t'}\|_{L_\infty} \leq \gamma_r$, which contradicts the inequality Eq. (60). Thus, the proof has been done.

Lemma 3. Given the system (42) and the L1 adaptive controller defined via Eqs. (47)–(49), the prediction error is bounded as.

$$\|\tilde{\omega}\|_{L_\infty} \leq \gamma_0 \quad (65)$$

Proof. The system prediction error dynamics can be obtained from Eq. (42) and Eq. (47) as

$$\dot{\tilde{\omega}}(\tau) = A_m \tilde{\omega}(\tau) - B \tilde{W}(\tau) H(\omega(\tau)) + B \varepsilon(\omega(\tau)) \quad (66)$$

where $\tilde{W}(\tau) = \hat{W}(\tau) - W$. Consider the following Lyapunov function candidate:

$$V(\tilde{\omega}(\tau), \tilde{W}(\tau)) = \tilde{\omega}^T(\tau) P \tilde{\omega}(\tau) + \tilde{W}^T(\tau) \Gamma^{-1} \tilde{W}(\tau) \quad (67)$$

Taking the derivative of Eq. (67), we can obtain

$$\begin{aligned} \dot{V}(\tilde{\omega}(\tau), \tilde{W}(\tau)) & = \dot{\tilde{\omega}}^T(\tau) P \tilde{\omega}(\tau) + \tilde{\omega}^T(\tau) P \dot{\tilde{\omega}}(\tau) \\ & + \dot{\tilde{W}}^T(\tau) \Gamma^{-1} \tilde{W}(\tau) + \tilde{W}^T(\tau) \Gamma^{-1} \dot{\tilde{W}}(\tau) \\ & = \tilde{\omega}^T(\tau) (A_m P + P A_m) \tilde{\omega}(\tau) - 2\tilde{\omega}^T(\tau) P B \tilde{W}(\tau) H(\omega(\tau)) \\ & + 2\tilde{\omega}^T(\tau) P B \varepsilon(\omega(\tau)) + 2\tilde{W}^T(\tau) \Gamma^{-1} \dot{\tilde{W}}(\tau) \\ & = -\tilde{\omega}^T(\tau) Q \tilde{\omega}(\tau) + 2\tilde{\omega}^T(\tau) P B \varepsilon(\omega(\tau)) \\ & - 2\tilde{\omega}^T(\tau) P B \tilde{W}(\tau) H(\omega(\tau)) \\ & + 2\tilde{W}^T(\tau) \text{Proj}(\tilde{W}(\tau) H^T(\omega(\tau)), \hat{W}(\tau)) \end{aligned} \quad (68)$$

Then from the projection operator characteristic, for any $\forall \tau \in [0, t]$

$$\dot{V}(\tau) \leq -\tilde{\omega}^T(\tau) Q \tilde{\omega}(\tau) + 2\tilde{\omega}^T(\tau) P B \varepsilon(\omega(\tau)) \quad (69)$$

Therefore, if $\|\tilde{\omega}(\tau)\| \geq 2\varepsilon^* \|PB\|/\lambda_{\min}(Q)$, then $\dot{V}(\tau) \leq 0$.

Projection operator ensures that $\tilde{W}(\tau) \in \Theta, \forall \tau \in [0, t]$, and therefore

$$\max_{\tau \in [0, t]} \tilde{W}^T(\tau) \Gamma^{-1} \tilde{W}(\tau) \leq W_{\max}/\Gamma_c, W_{\max} \triangleq \max_{W \in \Theta} 4\|W\|^2 \quad (70)$$

where $\mathbf{W}(\tau)$ is reshaped to a vector in the above projection operator analysis.

Thus, $\dot{\mathbf{V}}(\tau) \leq 0 (\tau \in [0, t])$ outside the compact set

$$\left\{ \|\tilde{\omega}(\tau)\| \leq \frac{2\varepsilon^* \|\mathbf{P}\mathbf{B}\|}{\lambda_{\min}(\mathbf{Q})} \right\} \cap \left\{ \|\tilde{\mathbf{W}}\| \leq \sqrt{W_{\max}} \right\} \quad (71)$$

This consequently implies that for any $\tau \in [0, t]$, we have

$$\|\tilde{\omega}\|_{L_\infty} \leq \gamma_0 \quad (72)$$

Theorem 1. *Given the system Eq. (42), the reference system Eqs. (57) and (58) and the L1 neural network adaptive controller defined via Eqs. (47)–(49) subject to condition (50), we have.*

$$\|\omega - \omega_{\text{ref}}\|_{L_\infty} \leq \gamma_1 \quad (73)$$

$$\|\mathbf{u}_{L1} - \mathbf{u}_{L1\text{-ref}}\|_{L_\infty} \leq \gamma_2 \quad (74)$$

Proof. First, we prove the bound (73) by a contradiction argument.

Since $\|\omega(0) - \omega_{\text{ref}}(0)\|_\infty = 0$ and both $\omega(t)$ and $\omega_{\text{ref}}(t)$ are continuous, there exists t' such that $\|\omega(t') - \omega_{\text{ref}}(t')\|_\infty > \gamma_1$ and $\|\omega(\tau) - \omega_{\text{ref}}(\tau)\|_\infty \leq \gamma_1 + \sigma$ for any $\tau \in [0, t']$. This leads to the following inequality:

$$\gamma_1 < \|\omega - \omega_{\text{ref}}\|_{L_\infty} \leq \gamma_1 + \sigma \quad (75)$$

From Eq. (47) and Eq. (58), one can write

$$\begin{aligned} \dot{\omega}(s) - \omega_{\text{ref}}(s) &= \bar{\mathbf{H}}(s)\varepsilon(s) + \bar{\mathbf{H}}(s)(\mathbf{C}(s) - \mathbf{I})(\hat{\eta}(s) - \eta(s)) \\ &= \bar{\mathbf{H}}(s)\varepsilon(s) + \bar{\mathbf{H}}(s)(\mathbf{C}(s) - \mathbf{I})(\eta_2(s) - \eta_3(s) + \eta_4(s)) \end{aligned} \quad (76)$$

where $\eta_2(s), \eta_3(s)$ and $\eta_4(s)$ are the Laplace transformations of signals $\hat{\mathbf{W}}(t)\mathbf{H}(\omega(t))$, $\mathbf{W}(t)(\mathbf{H}(\hat{\omega}(t)) - \mathbf{H}(\omega(t)))$ and $\mathbf{W}(t)(\mathbf{H}(\hat{\omega}(t)) - \mathbf{H}(\omega_{\text{ref}}(t)))$, respectively.

Then from Eq. (66), we have

$$\tilde{\omega}(s) = \bar{\mathbf{H}}(s)(-\eta_2(s) + \varepsilon(s)) \quad (77)$$

Eq. (76) can be rewritten as

$$\begin{aligned} \dot{\omega}(s) - \omega_{\text{ref}}(s) &= \bar{\mathbf{H}}(s)\varepsilon(s) \\ &\quad + \bar{\mathbf{H}}(s)(\mathbf{C}(s) - \mathbf{I})(-\eta_3(s) + \eta_4(s) + \varepsilon(s)) \\ &\quad - (\mathbf{C}(s) - \mathbf{I})\tilde{\omega}(s) \end{aligned} \quad (78)$$

Using the RBF approximation in Eq. (45), one arrives at

$$\|\mathbf{W}(\mathbf{H}(\hat{\omega}(\tau)) - \mathbf{H}(\omega(\tau)) - (\mathbf{f}(\hat{\omega}(\tau)) - \mathbf{f}(\omega(\tau))))\|_\infty \leq 2\varepsilon^* \quad (79)$$

for $\tau \in [0, t']$, the relationship of Eqs. (43) and (44) implies that

$$\begin{aligned} \|\eta_{3t'}\|_{L_\infty} &= \|\mathbf{W}(\mathbf{H}(\hat{\omega}) - \mathbf{H}(\omega))\|_{L_\infty} \\ &\leq \|\mathbf{f}(\hat{\omega}) - \mathbf{f}(\omega)\|_{L_\infty} + 2\varepsilon^* \\ &\leq L\|\tilde{\omega}_{t'}\|_{L_\infty} + 2\varepsilon^* \end{aligned} \quad (80)$$

Similarly

$$\|\eta_{4t'}\|_{L_\infty} \leq L\|\hat{\omega} - \omega_{\text{ref}}\|_{L_\infty} + 2\varepsilon^* \quad (81)$$

Lemma 1 leads to the following upper bound:

$$\begin{aligned} \|\hat{\omega} - \omega_{\text{ref}}\|_{L_\infty} &\leq \|\mathbf{G}(s)\|_{L1} (L\|\hat{\omega} - \omega_{\text{ref}}\|_{L_\infty} \\ &\quad + L\|\tilde{\omega}_{t'}\|_{L_\infty} + 5\varepsilon^*) + \|\mathbf{C}(s) - \mathbf{I}\|_{L1} \|\tilde{\omega}_{t'}\|_{L_\infty} + \|\bar{\mathbf{H}}(s)\|_{L1} \varepsilon^* \end{aligned} \quad (82)$$

Using the condition defined in Eq. (50), we have

$$\|\hat{\omega} - \omega_{\text{ref}}\|_{L_\infty} \leq \frac{\|\mathbf{G}(s)\|_{L1} (L\|\tilde{\omega}_{t'}\|_{L_\infty} + 5\varepsilon^*) + \|\mathbf{C}(s) - \mathbf{I}\|_{L1} \|\tilde{\omega}_{t'}\|_{L_\infty} + \|\bar{\mathbf{H}}(s)\|_{L1} \varepsilon^*}{1 - \|\mathbf{G}(s)\|_{L1} L} \quad (83)$$

Since $\|\omega - \omega_{\text{ref}}\|_{L_\infty} \leq \|\hat{\omega} - \omega_{\text{ref}}\|_{L_\infty} + \|\tilde{\omega}_{t'}\|_{L_\infty}$, it follows from inequality Eq. (72) that $\|\omega - \omega_{\text{ref}}\|_{L_\infty} \leq \gamma_1$, which contradicts the condition Eq. (75). Hence inequality Eq. (73) holds. It follows from Eq. (49) and Eq. (57) that

$$\begin{aligned} \mathbf{u}_{L1}(s) - \mathbf{u}_{L1\text{-ref}}(s) &= \mathbf{C}(s)\hat{\eta}(s) - \mathbf{C}(s)\eta(s) \\ &= \mathbf{C}(s)(\bar{\mathbf{W}}(s)\mathbf{H}(\omega(s)) - \mathbf{W}(s)\mathbf{H}(\omega_{\text{ref}}(s))) \\ &= \mathbf{C}(s)(\bar{\mathbf{W}}(s) - \mathbf{W}(s))\mathbf{H}(\omega(s) + \mathbf{C}(s)\mathbf{W}(s)(\mathbf{H}(\omega(s)) - \mathbf{H}(\omega_{\text{ref}}(s)))) \\ &= \mathbf{C}(s)\eta_2(s) + \mathbf{C}(s)\mathbf{W}(s)(\mathbf{H}(\omega(s)) - \mathbf{H}(\omega_{\text{ref}}(s))) \end{aligned} \quad (84)$$

From Eq. (80), we obtain

$$\begin{aligned} \|\mathbf{W}(s)(\mathbf{H}(\omega(s)) - \mathbf{H}(\omega_{\text{ref}}(s)))\|_{L_\infty} \\ \leq L\|\omega - \omega_{\text{ref}}\|_{L_\infty} + 2\varepsilon^* \end{aligned} \quad (85)$$

From Lemma 4 in Ref. 21, we know that if $(\mathbf{A}, \mathbf{b})$ is controllable and $(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{b}$ is strictly proper and stable, there exists $\mathbf{c}_o \in \mathbb{R}^n$ such that the transfer function $\mathbf{c}_o^T(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{b}$ is minimum phase with relative degree one. $\mathbf{C}(s)\eta_2(s)$ can be rewritten, according to inequality (81), as

$$\begin{aligned} \mathbf{C}(s)\eta_2(s) &= \mathbf{C}(s)\frac{1}{\mathbf{c}_o^T \mathbf{H}(s)} \mathbf{c}_o^T \bar{\mathbf{H}}(s)\eta_2(s) \\ &= \mathbf{C}(s)\frac{1}{\mathbf{c}_o^T \bar{\mathbf{H}}(s)} \mathbf{c}_o^T (-\tilde{\omega}(s) + \bar{\mathbf{H}}(s)\varepsilon(s)) \\ &= -\mathbf{C}(s)\frac{1}{\mathbf{c}_o^T \bar{\mathbf{H}}(s)} \mathbf{c}_o^T \tilde{\omega}(s) + \mathbf{C}(s)\varepsilon(s) \end{aligned} \quad (86)$$

Since $\mathbf{C}(s)$ is stable and strictly proper, the complete system $\mathbf{C}(s)\frac{1}{\mathbf{c}_o^T \bar{\mathbf{H}}(s)}$ is proper and stable, which implies that its L1 gain exists and is finite. Hence, we have

$$\|\mathbf{C}(s)\eta_2(s)\|_{L_\infty} \leq \left\| \mathbf{C}(s)\frac{1}{\mathbf{c}_o^T \bar{\mathbf{H}}(s)} \mathbf{c}_o^T \right\|_{L1} \|\tilde{\omega}(s)\|_{L_\infty} + \|\mathbf{C}(s)\|_{L1} \varepsilon^* \quad (87)$$

which leads to the following upper bound:

$$\begin{aligned} \|\mathbf{u}_{L1}(s) - \mathbf{u}_{L1\text{-ref}}(s)\|_{L_\infty} &\leq \left\| \mathbf{C}(s)\frac{1}{\mathbf{c}_o^T \bar{\mathbf{H}}(s)} \mathbf{c}_o^T \right\|_{L1} \|\tilde{\omega}(s)\|_{L_\infty} \\ &\quad + L\|\mathbf{C}(s)\|_{L1} \|\omega - \omega_{\text{ref}}\|_{L_\infty} \\ &\quad + 3\|\mathbf{C}(s)\|_{L1} \varepsilon^* = \gamma_2 \end{aligned} \quad (88)$$

4. Simulation

In this section, the control architecture designed before is verified through different simulation subjects. The simulation environment is given first, and the corresponding simulation results are then analyzed.

4.1. Simulation environment and controller parameters

The aerodynamic force and moment rely on the propeller slipstream to a large extent as discussed in Section 2. The slipstream areas in different regions should be considered and the nominal condition is listed in Table 5.

The areas in different regions are illustrated in Table 5 under nominal conditions which means that the tail-sitter UAV is in the hover flight without any movement. The slipstream area changes with the propeller thrust, distance behind the propeller disc and induced velocity near the propeller disc in the simulation, as described in Eq. (7) and Eq. (8).

The simulation initial values are given in Table 6.

According to our previous embedded algorithm implementation experience, the following factors should be considered:

- (1) The stability margin of the system. Although this may not be intuitive in some control algorithms, this determines how aggressive we can tune the control parameters. The pure time delay part within the system would significantly hurt the stability margin.
- (2) The limited algorithm running frequency in flight control hardware should be noticed. For an L1 adaptive controller with continuous form, the maximal value of adaptation rate is restricted to the running frequency of the attitude control loop.
- (3) The UAV's practical physical performance should be considered when choosing control parameters. Excessive pursuit of control performance while ignoring the UAV's practical physical performance will eventually lead to control performance degradation.

Table 5 Areas within and outside propeller slipstream.

Parameter	Description	Value
S_{in_w} (m ²)	Area within slipstream (wing)	0.0494
S_{out_w} (m ²)	Area out of slipstream (wing)	0.1119
S_{in_de} (m ²)	Area within slipstream (control surface)	0.0116
S_{out_de} (m ²)	Area out of slipstream (control surface)	0.0396

Table 6 Tail-sitter UAV's simulation initial values.

Parameter	Description	Value
$[X_{e0}, Y_{e0}, Z_{e0}]$ (m)	Position in inertial axes	$[0, 0, 0]$
$[u_0, v_0, w_0]$ (m/s)	Velocity in body axes	$[0, 0, -0.5]$
$[p_0, q_0, r_0]$ (°/s)	Body rotation rates	$[0, 0, 0]$
$[\hat{p}_0, \hat{q}_0, \hat{r}_0]$ (°/s)	Angular rates of state predictor in L1 adaptive controller	$[0, 0, 0]$
$[\phi_0, \theta_0, \psi_0]$ (°)	Euler angle	$[0, 0, 0]$
δ_e (°)	Control surface deflection	$[-6, -6]$

Furthermore, the trade-off between the computational burden and the control accuracy should also be considered. If the computational burden of the algorithm is too large, it will inevitably lead to a decrease in the algorithm's running frequency for a specific flight control hardware. Low control algorithm running frequency has a detrimental effect on control performance. If the control precision cannot be violated, there are three ways to cope with it: (A) reducing the algorithm's computational complexity in a certain way; (B) using a flight control hardware with more computing power; (C) seeking other control algorithms that need less computation.

The following control parameter and simulation parameters are selected based on the practical considerations described above.

According to our previous studies, the dynamics of the control surface and motor can be modeled as $1/(0.03s + 1)$ and $1/(0.02s + 1)$, respectively. Furthermore, the original data discrete sampling, actuators' time delay and algorithm's limited calculation frequency under real hardware would all bring time delay to system input. According to our previous studies, the controller's time delay margin should be larger than 20 ms to offer sufficient robustness. In the following simulation, the system input time delay is set 25 ms for more rigorous tests. To make the simulation results more reliable, the maximal control efforts (the maximal control surface and deflection and motor thrust) are set as shown in Table 1. The control constraints are guaranteed by setting upper and lower bound of the control surface and motor thrust during simulation. There are two situations that the desired control may exceed the actuators' maximal values: (A) If the desired controls exceed the UAV's ability when UAV encounters overlarge disturbances, the performance degradation is inevitable; (B) If the desired controls exceed actuators' maximal value because of saturation, the anti-windup strategy could be used to guarantee the control performance.

Much time is spent in tuning the cascade PID controller and the parameters are given in Table 7.

The compensation for system uncertainties and environmental disturbances is realized by RBFN in L1 adaptive control. The neural network parameters are obtained by trial-and-error methods. The number of nodes in the network input layer, hidden layer and output layer are set as 3, 5 and 3, respectively, as shown in Fig. 8. The center vector for neuron is defined as $\mathbf{z} = [-1, -0.5, 0, 0.5, 1; -1, -0.5, 0, 0.5, 1; -1, -0.5, 0, 0.5, 1]$ and the width of radial basis is set to 2. From the actual parameter tuning process, the changes of RBFN coefficients within a certain range would not bring a noticeable impact on the estimation of uncertainties and disturbances.

Table 7 PID control parameters.

Parameter	Roll	Pitch	Yaw
P_Ω	6	4.5	3.8
P_ω	0.12	0.3	0.22
I_ω	0.002	0.002	0.02
D_ω	0.003	0.003	0

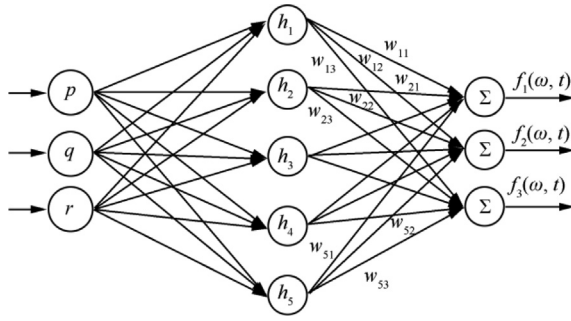
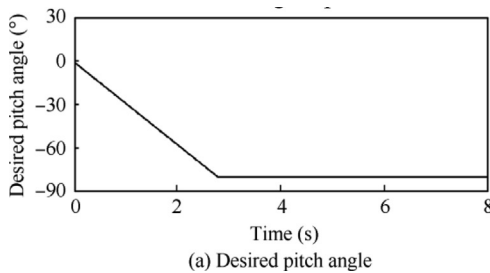


Fig. 8 Illustration of RNFN in L1 adaptive control architecture.

A large adaptive gain helps to decrease the error dynamics deviation, and with the help of a low-pass filter, the system robustness would not hurt too much, however, larger adaptive gain requires higher computing efforts of the flight control hardware and the adaptive gain is set as 400. The bandwidth of the low-pass filter in the L1 adaptive controller is set as $15/(s + 15)$ to trade off between system performance and robustness. The desired dynamics of the angular rate loop are set as $-10\mathbf{I}_{3 \times 3}$ to obtain a quick angular rate tracking in this study. Although larger number setting of desired angular rate dynamics would make the UAV track angular rate faster in theory, the UAV's limited physical ability (finite actuator bandwidth, the UAV's moment of inertia, etc.) may not be able to follow overly aggressive control expectation.

In the simulation analysis, the desired transition trajectory should be given first. The different settings of desired pitch angles and desired speeds have a substantial impact on the transition result. The transition strategy using optimization method has been researched in our previous studies. In that study, an optimal function and constraints are defined, then gradient descent algorithm is used to obtain the desired pitch angles and speeds. Shorter transition time settings for pitch angles and speeds would lead to drastic changes in aerodynamic characteristics such as the variations of angle of attack and dynamic pressure, therefore, more control manipulations are needed. The linear transition trajectory is given as reference values because the focus of this article is control system design. The influence of different time settings between desired pitch angles and speeds under linear transition trajectory will be discussed in Section 4.2.3.

The transition finish time is set to 3 s moderately to test the control system in this study. The desired transition pitch angle and desired transition flight speed in the Z_b axis of the body coordinates are illustrated as Fig. 9.



4.2. Simulation tests under different conditions

4.2.1. Nominal system without disturbance and uncertainty

In this simulation, the control system attitude tracking performance is tested in the nominal system without any environmental disturbances or system uncertainties. Comparisons are made between classical single-loop PID control, cascade PID and the algorithms proposed in this study. The single-loop PID uses errors between the desired attitude and real attitude as control input and outputs the desired control moment to actuators. The attitude expression is based on the Euler angle, therefore, the final desired pitch angle is set to -80° to avoid singularity. The simulation results are shown in Fig. 10. In Fig. 10, CMD represents the desired pitch angle during the transition process, L1 represents the pitch angle tracking result proposed in this study, PID(C) and PID(S) represent the pitch angle tracking results of the single-loop PID and cascade PID controllers, respectively.

From Fig. 10, all three control methods could make the tail-sitter UAV finish transition in the nominal system. However, the single-loop PID controller shows the worst performance, and large tracking errors occur during the transition process. The fluctuation is also the largest among these three methods after transition. Cascade PID control shows great

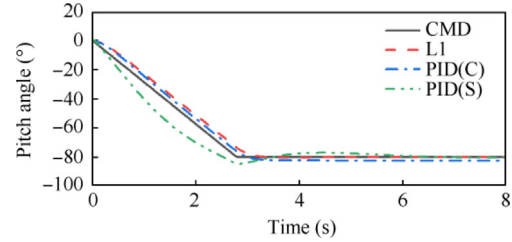


Fig. 10 Pitch angle tracing in nominal system.

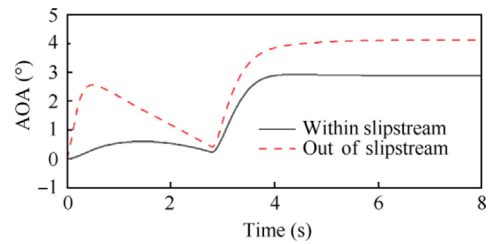


Fig. 11 Angle of attack.

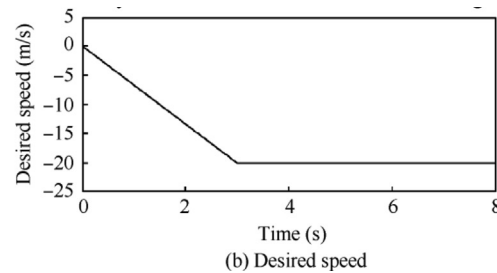


Fig. 9 Desired transition set point.

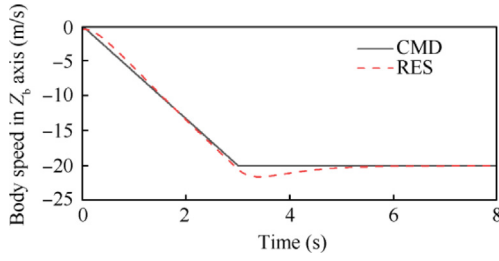


Fig. 12 Speed control.

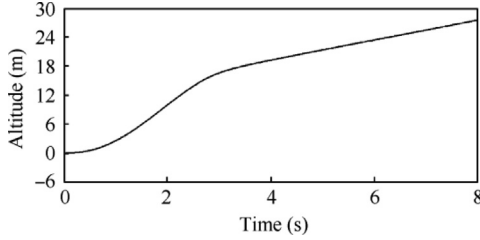


Fig. 13 Altitude change.

improvement and many of the tracking errors during the transition process are eliminated. However, the tracking errors are not as small as those of the L1 neural network adaptive controller in the level flight phase. In general, the control performance of cascade PID control and L1 neural network adaptive control show the same level under the nominal system.

The Angle of Attack (AOA) within and outside the propeller slipstream, flight speed and altitude change during the transition process are shown in Fig. 11, Fig. 12 and Fig. 13, respectively. The results from the L1 neural network adaptive control system are displayed, where RES represents the real body speed in Z_b axis.

The speed control used here is a standard PID controller. Given the tail-sitter UAV coordinate definition, -20 m/s in Fig. 12 actually means that the forward speed is 20 m/s. From Fig. 12, the tail-sitter UAV reaches the desired flight speed after the transition and overshoots exist at the end of the transition. The overshoot is less than 2 m/s and the results are acceptable. From Fig. 11 and Fig. 13, it can be observed that the angles of attack within and outside the propeller slipstream are relatively small, which would lead to altitude climbing. The reason is that small angles of attack would generate sufficient lift for the tail-sitter UAV. However, remarkably, this small angle of attack and altitude climbing are related to the transition strategy, which will be discussed in Section 4.2.3.

4.2.2. Complete system with disturbances and uncertainties

In this part, the control performances of different controllers are tested under disturbances and uncertainties. Furthermore, the transition time is set to 2 s for more rigorous test. The step signal and low-frequency sinusoidal signal are mainly used to verify the attitude tracking performance of the tail-sitter UAV. Extra disturbances are added to the pitching channel to test the pitch tracking ability which is crucial to the transition process. The following triaxial superimposed moments are added to the system input as shown in Table 8.

Moreover, wind disturbances are also included in the analysis and they will bring changes in dynamic pressure and angle of attack to the tail-sitter UAV aerodynamics. The wind model is established according to the MIL-F-8785C standard, as mentioned in Section 2.3.5. The wind fields are shown in Fig. 14 and have been transferred to the body coordinates.

Considering that the performance of single-loop PID is the worst among these three control architectures, comparisons are mainly made between cascade PID control and the L1 neural network adaptive control system.

The angles of attack within and outside the propeller slipstream under disturbances are shown in Fig. 15. It can be observed that there exist more drastic fluctuations compared to the nominal system without disturbances, which would impact the tail-sitter UAV's aerodynamic force and moment.

The triaxial Euler angle tracking performances are given in Fig. 16. In Fig. 16, CMD represents the desired Euler angle, PID(C) represents the attitude tracking performance using the cascade PID controller, L1 represents the attitude tracking performance using the method proposed in this study.

The outputs of the cascade PID controller and L1 neural network adaptive controller under the control architecture of Fig. 7 are shown as Fig. 17 and the contribution of these two controllers can be seen clearly.

The Maximal Absolute Error (MAE) and Integral Squared Error (ISE) are used to evaluate different control effects. MAE is defined as $\max |e|$ and ISE is defined as $\int_{t_0}^{t_f} e^2(t)dt$. The cor-

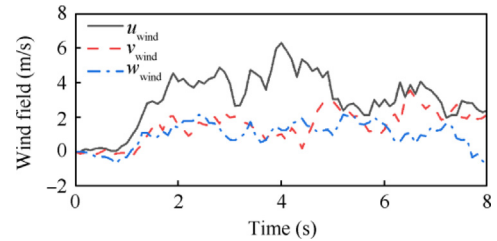


Fig. 14 Linear speed in wind field.

Table 8 Triaxial disturbance moments.

Channel	Moment
Roll	$M_1 = 0.1 \sin(0.5t), M_2 = 0.05 \sin(3t), M_3 = \begin{cases} 0.1 & 0.5 \leq t \leq 1.5 \\ 0 & t < 0.5, t > 1.5 \end{cases}$
Pitch	$M_1 = 0.1 \sin(0.5t), M_2 = 0.25 \sin(3t), M_3 = \begin{cases} 0.1 & 0.5 \leq t \leq 1.5 \\ 0 & t < 0.5, t > 1.5 \end{cases}, M_4 = \begin{cases} 0.3 & 0 \leq t \leq 2.5 \\ 0 & t > 2.5 \end{cases}$
Yaw	$M_1 = 0.1 \sin(0.5t), M_2 = 0.05 \sin(3t), M_3 = \begin{cases} 0.1 & 0.5 \leq t \leq 1.5 \\ 0 & t < 0.5, t > 1.5 \end{cases}$

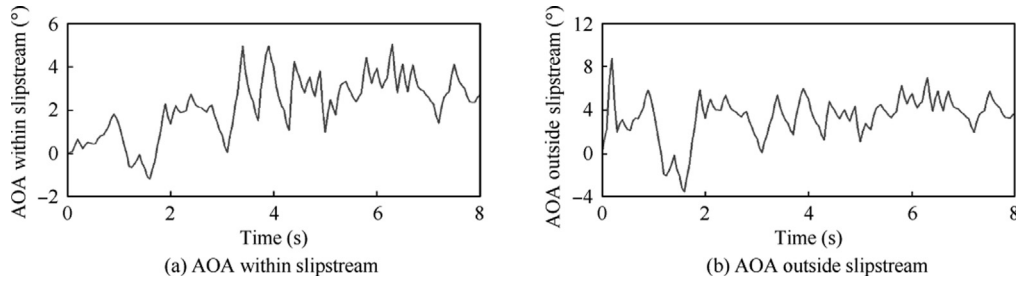


Fig. 15 Angle of attack changes during transition process.

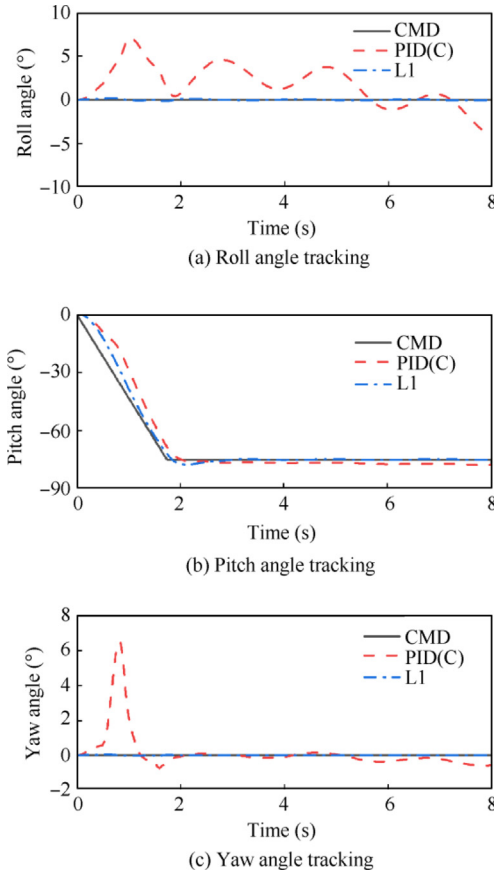


Fig. 16 Triaxial Euler angle tracking performance.

responding computed results are listed in [Tables 9 and 10](#), respectively.

By using a cascade PID controller, the maximal absolute error in roll and yaw control are 7.2053° and 6.45° respectively while the errors in the control system that we propose here are 0.1924° and 0.776° respectively. Smaller angle errors in roll and yaw angle would maintain UAV smooth and steady during transition process. Furthermore, visible errors of PID controller could also be observed after transition. The MAE in the pitch tunnel of PID controller is 17.2587° and that of L1 neural network controller is 9.4814° . The relatively poor control precisions of PID controller increase the risk of safety transition. The MAE could reflect the transient performance during transition while the investigation of control process

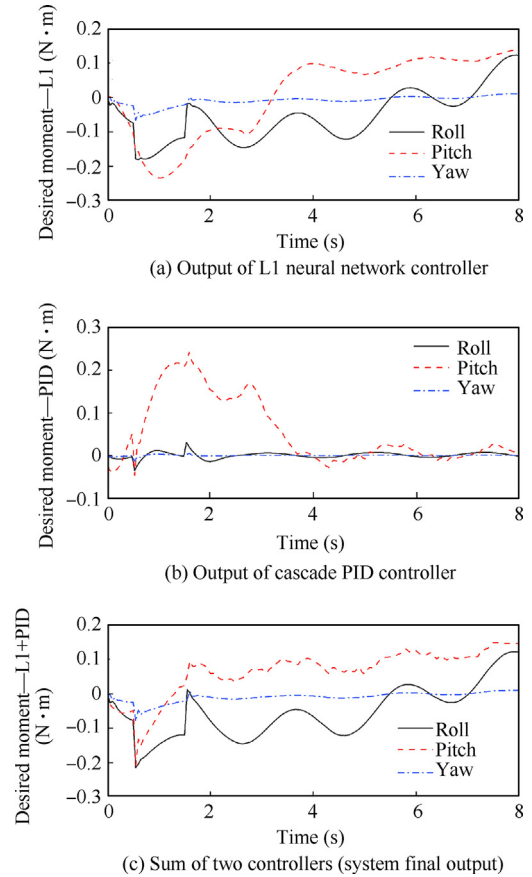


Fig. 17 Desired control moment generated by controllers.

Table 9 Control performance evaluation (PID).

Error	Roll	Pitch	Yaw
MAE	7.2053	17.2587	6.45
ISE	66.2854	256.4521	10.3439

can be evaluated by ISE. Larger ISE means greater fluctuations during transition process. From the ISE results in [Table 9](#) and [Table 10](#), better process control can also be observed in the L1 neural network control system. In conclusion, the L1 neural network adaptive control system proposed in this study

Table 10 Control performance evaluation (L1 neural network).

Error	Roll	Pitch	Yaw
MAE	0.1924	9.4814	0.776
ISE	0.351	66.5026	0.15

could attenuate the disturbances considerably and enhance flight safety compared to single use of cascade PID controller.

The changes in parameters related to the neural network are shown in Fig. 18.

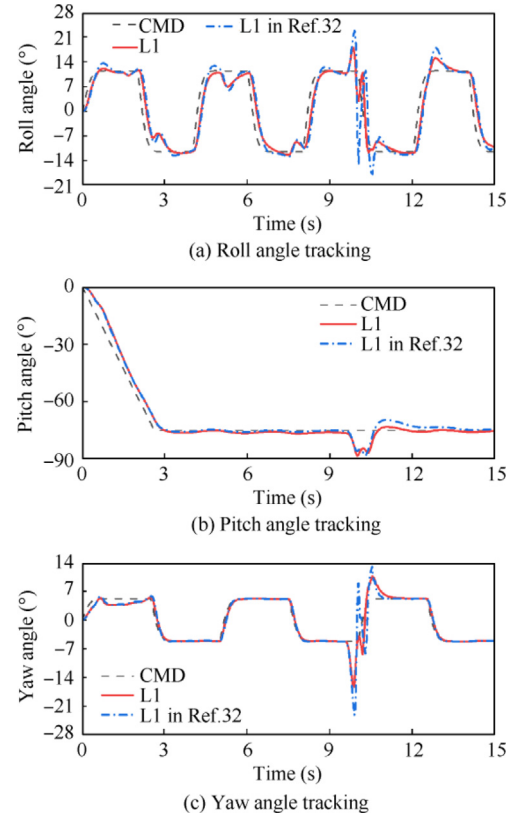
Furthermore, the control performances are also compared between the method proposed in this study and the method conducted in Ref. 32. In Ref. 32, the adaptation law is the product of adaptive gain and state errors between the real system and predictor. The mathematical description of the adaptation law is shown as

$$\dot{\hat{\eta}} = -\Gamma(\hat{x}(t) - x(t))$$

where $\hat{x}(t)$ is the state of the predictor, $x(t)$ is the state of the real system, Γ is the adaptation gain.

The simulation results using these two control architectures are shown in Fig. 19. The control performance of these two L1 adaptive controls is almost the same when the adaptation gain is chosen to be relatively large through early simulation. From our previous experience, the algorithm operating frequency, flashed into Pixhawk1 hardware, is between 200 Hz and 250 Hz and the maximal adaptation gain should not be larger than 160, otherwise, unexpected performance would occur, which means that the tail-sitter UAV's attitude starts shaking. Therefore, the adaptation gains are both set to 160 to compare the control effect of the two L1 adaptive control architectures. In Fig. 19, CMD represents the desired attitude, L1 represents the method proposed in this study, L1 in Ref. 32 represents the method proposed in Ref. 32.

Although the control architecture proposed in Ref. 32 could obtain reasonable results, the tracking precision is still not as good as the controller proposed in this study. The complex desired attitude shown in Fig. 18 leads the maximal roll angle tracking error to 11° and 25° by using the method proposed in this study and the method proposed in Ref. 32, respectively. Furthermore, the method in this study also shows better yaw angle tracking performance with the maximal absolute error of 11° while the tracking error using method in Ref. 32 is 18°. The reason is that the adaptation law in Ref. 32 is just a product between the adaptation gain and state errors. It is generally a linear integral of the state errors. However, the adaptation law in this study uses the good approximation ability of RBFN, which is a nonlinear function, to estimate the

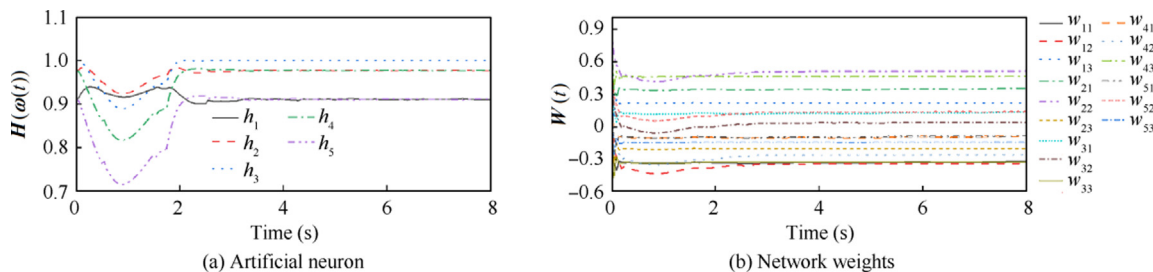
**Fig. 19** Comparison between L1 adaptive controllers.

disturbances and uncertainties. More approximate and accurate estimations could be obtained and then be used for control output. Therefore, a better control precision could be obtained.

Furthermore, the influences of transition time setting on the transition trajectory are also discussed.

4.2.3. Different transition time setting analysis

In the previous simulations, the desired pitch angle and desired flight speed were set as a linear relation with time during the transition process. The time durations of the desired pitch angle and desired flight speed are the same. Although there exist studies that compute a nonlinear transition strategy to guarantee relatively small altitude changes based on optimization ideas, the different time settings of these desired values under linear transition strategy are rarely analyzed and will be discussed in this section. To avoid the influences of disturbances and uncertainties, simulations are conducted based on

**Fig. 18** Parameters related to neural network.

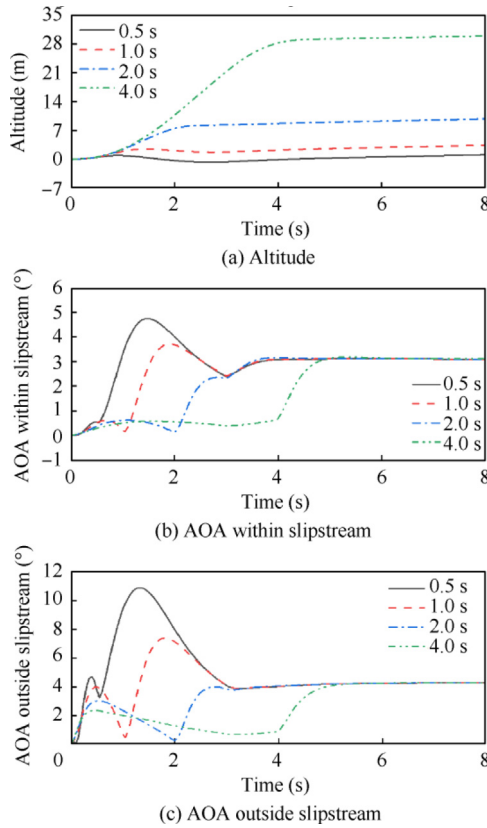


Fig. 20 Relation between transition trajectory and time setting (flight speed time fixed).

the nominal system which does not contain disturbances and uncertainties.

The time of the desired flight speed during the transition process is set to 3 s and the time of the desired pitch angle is set to 0.5, 1, 2, 4 s. The changes in altitude and angles of attack are investigated as Fig. 20.

When the time setting of the desired transition flight speed is fixed, less time setting of the desired pitch angle would bring a smaller altitude change. When the desired pitch angle time setting is 0.5 s, the altitude change during transition is almost 0. However, it is worth noting that a transition time setting that is too small would lead to more drastic angle of attack changes. Furthermore, from Figs. 20 (b) and (c), the influences of transition pitch angle time setting on the angle of attack changes can also be observed. In general, a longer transition pitch angle time setting would also lead to greater height variation under the linear transition strategy.

Then the desired transition pitch angle time is fixed to 3 s, and the time of the desired flight speed is set to 0.5, 1, 2, 4 s. The changes in altitude and angles of attack during the transition process are discussed as Fig. 21.

When the desired transition pitch angle time is fixed, the shorter the desired flight speed time is set, the higher the tail-sitter UAV climbs. The reason is that a relatively short time setting of the desired flight speed would make the tail-sitter UAV reach the cruising speed quickly, and there still exists a relatively large pitch angle at this moment which leads to obvious altitude climbing. Compared to the results from a fixed transition flight speed time, it would be more efficient to

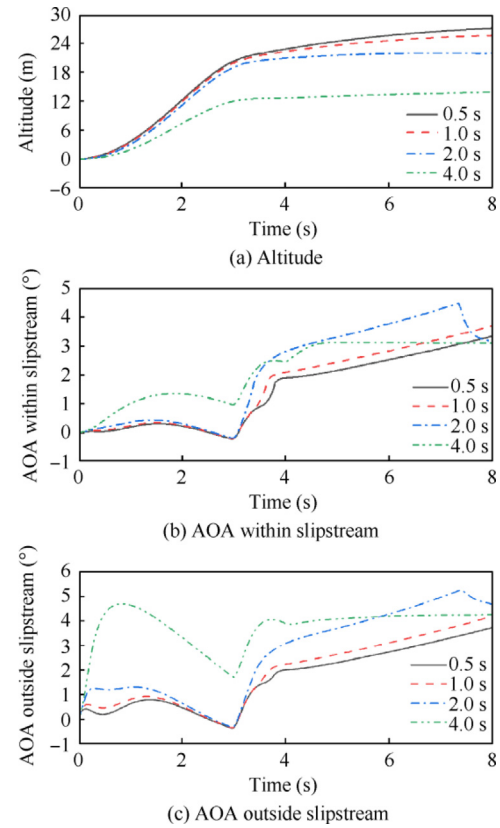


Fig. 21 Relation between transition trajectory and time setting (pitch angle time fixed).

change the transition altitude variation by changing the desired pitch angle time setting.

Based on the above analysis, transition altitude planning would be effectively realized by adjusting the time difference between the transition time setting of the desired pitch angle and desired flight speed. This provides another idea to cope with transition altitude variation instead of complicated non-linear optimization solutions. However, it should be noted that the conclusion obtained in this paper can only be applied to the tail-sitter UAV of this study because different UAVs have different aerodynamic characteristics. However, the analysis method mentioned above can be extended to other types of tail-sitter UAVs.

5. Conclusions

The key features of the tail-sitter UAV during the transition process are modeled first in this paper, such as aerodynamic characteristics within full angles of attack, and the influences of the propeller slipstream on the dynamic pressure and angle of attack in the different parts of wings and control surfaces. These established models are the research foundation for the control system verification.

A novel control architecture of L1 neural network adaptive control system associated with cascaded PID control is developed to improve the compensation ability during the transition process and guarantee the security transition. The L1 neural network adaptive control algorithm is revised to cope with multi-input multi-output system and the corresponding

theoretical analyses are given. Numerical simulation results show the effectiveness of the proposed control method. Using the evaluation criteria of MAE and ISE for transition process analysis, the pitch angle tracking of the method proposed in this study has at least twice the tracking precision of cascaded PID control. The roll and yaw control results show more obvious improvement of control effect. Furthermore, higher control precision and smaller overshoots could also be guaranteed by the method proposed in this study compared to the L1 control method proposed in the reference which uses a proportional adaptation law.

Finally, the influences of the transition time setting between the desired pitch angle and flight speed on the transition altitude variation are discussed. This raises a new idea to cope with the transition altitude change under a linear transition strategy instead of complicated nonlinear transition strategy computation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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